Finite Euler reflection rigidity and a positive Poisson boundary

Paper C: Theorem Forge II

Observational SUCC research dossier

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Abstract

A finite quotient of Euler factors with real bases greater than one cannot satisfy reflection about a nonzero real center unless its two base multisets are identical. Equal cardinality is a conclusion, not an assumption. We give an exact reflection failure for a density-one Beurling counterfeit. For infinite sources we prove a separate sufficient theorem: a positive tempered measure with a normalized atom at zero, a unit gap and Fourier self-duality is the integer Dirac comb. An explicit contour-decay hypothesis converts the usual Gaussian-completed functional equation into this rigidity criterion, including for generalized real exponents. No growth estimate is inferred from a formal functional equation. These are source-identification results, not a proof of the Riemann hypothesis. Scholarly novelty remains unverified.

1 Contract and source boundary

The finite theorem concerns finite multisets of real numbers greater than one, with multiplicities retained. The infinite theorem concerns positive source measures and states all analytic hypotheses separately. The earlier repository result ASM-01 constructs, beyond every finite observation horizon, a collision-free Beurling source with density one, the ordinary prime number theorem, residue one and the same critical-strip zeta divisor as ordinary arithmetic, but a failed additive-rank braid. Its finite Euler quotient is holomorphic and nonzero on the right half-plane; it does not preserve the completed functional equation. We identify a precise extra condition that excludes this family and a sufficient infinite-source extension.

The base is PR179, commit `81684b4b8956f9f5da56c376ba7e4339277eb665`; the predecessor proof is `research/2026-10-18/ARITHMETIC_SOURCE_MINIMALITY.md`. Nothing below assumes that sibling PR178 has been merged. No zero ordinate is an input to a construction or a test.

2 Finite factors: no cardinality assumption

For finite multisets $U, V \subset (1, \infty)$ define the meromorphic function

$$F(s) = \frac{\prod_{u \in U}(1 - u^{-s})}{\prod_{v \in V}(1 - v^{-s})}, \qquad u^{-s} = e^{-s \log u}.$$

Empty products equal one. Factors and their multiplicities are literal; coincident powers of different bases are not identified.

Theorem 1 (Finite reflection rigidity)**.** *Let $c \in \mathbb{R} \setminus \{0\}$. If $F(s) = F(c - s)$ as meromorphic functions on $\mathbb{C}$, then $F \equiv 1$ and $U = V$ as multisets. In particular the statement holds for $c = 1$, without assuming $|U| = |V|$.*

Proof. Put $\epsilon = (-1)^{|U|+|V|}$ and $C = \prod_{u\in U} u / \prod_{v\in V} v > 0$. The elementary factor identity $1 - u^x = -u^x(1 - u^{-x})$ gives, for real t sufficiently large,

$$F(c - t) = \epsilon C^{t-c} F(t - c). \tag{1}$$

Both $F(t)$ and $F(t - c)$ are strictly positive and tend to one as $t \to \infty$. Reflection and (1) first imply $\epsilon = 1$, and then $C^{t-c} = F(t)/F(t - c) \to 1$, so $C = 1$. Consequently $F(t) = F(t - c)$ for all sufficiently large t. Whether c is positive or negative, this gives $F(t) = F(t + n|c|)$ for every $n \geq 0$ and every sufficiently large t. Taking $n \to \infty$ proves $F(t) = 1$. The meromorphic identity theorem gives $F \equiv 1$.

Cancel equal bases on the two sides with their full multiplicities. If any base remains, let b be the least one and let $m_b \neq 0$ be its signed multiplicity (numerator minus denominator). On the positive real axis the real logarithm is defined and

$$\log F(t) = \sum_a m_a \log(1 - a^{-t}), \qquad \lim_{t\to\infty} b^t \log F(t) = -m_b.$$

Indeed $\log(1-z) = -z+O(z^2)$ as $z \to 0$; every remaining base $a > b$ contributes a quantity tending to zero after multiplication by b^t, and the sum is finite. This contradicts $F \equiv 1$. Thus no base remains. □

There is also a direct cardinality check when $c = 1$. Each factor has a simple zero at zero, so $\operatorname{ord}_0 F = |U| - |V|$. Since $F(1)$ is finite and strictly positive, reflection forces this order to vanish. This independently handles unequal counts. Possible poles on the imaginary axis cause no defect in the proof: its real-axis ratios have no zero or pole, and all other locations are handled by meromorphic continuation.

Remark 2 (Sharp excluded center). For $U = \{2, 6\}$ and $V = \{3, 4\}$, the counts and products agree and (1) gives $F(s) = F(-s)$. The quotient is nonconstant by the least-base argument. Thus the restriction $c \neq 0$ is necessary. Equal counts and equal products alone are insufficient at $c = 1$.

2.1 An exact counterfeit rejected by completion

Delete ordinary primes $p = 3, q = 11$ and introduce primitive values $a = 9/2, b = 77/17$. Retain every other ordinary prime and count formal commutative words. The rational valuations at the deleted primes are

$$(v_3(a), v_{11}(a)) = (2, 0), \qquad (v_3(b), v_{11}(b)) = (0, 1),$$

so distinct words have distinct values. For $\Re s > 1$ its zeta function is $Z_B(s) = \zeta(s)F(s)$ with $U = \{3, 11\}$ and $V = \{9/2, 77/17\}$. The identity

$$(1 - a^{-1})(1 - b^{-1}) = (1 - 3^{-1})(1 - 11^{-1})$$

gives $F(1) = 1$. None of these factors vanishes in $\Re s > 0$, so the same critical-strip divisor and residue one follow from ordinary zeta there. But exact rational evaluation yields

$$F(2) = \frac{504}{517}, \qquad F(-1) = \frac{34}{21}. \tag{2}$$

In particular the same Gaussian completion cannot obey reflection. More generally, if $\Lambda_B = \Lambda_\zeta F$ and both completed functions obey reflection, meromorphic cancellation of the nonzero function Λ_ζ forces $F(s) = F(1 - s)$. Theorem 1 then forbids every nontrivial finite alteration of primitive bases, including alterations with repeated generators or coincident powers.

3 A positive Poisson rigidity lemma

Use the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi ix\xi}\,dx$ and its extension to tempered distributions. Write $\delta_{a\mathbb{Z}} = \sum_{k\in\mathbb{Z}} \delta_{ak}$.

Theorem 3 (Unit gap and positive self-duality)**.** *Let μ be a positive tempered Radon measure on $\mathbb{R}$ with*

$$\mu(\{0\}) = 1, \qquad \mu((-1,1)\setminus\{0\}) = 0, \qquad \widehat{\mu} = \mu.$$

Then $\mu = \delta_{\mathbb{Z}}$. Equivalently, an arbitrary positive mass ρ at zero gives $\mu = \rho\delta_{\mathbb{Z}}$ under the other two hypotheses.

Proof. Set

$$\phi(x) = \left(\frac{\sin \pi x}{\pi x}\right)^2, \quad \phi(0) = 1, \qquad T(\xi) = (1 - |\xi|)_+.$$

The Fourier identity $\widehat{\phi} = T$ follows by squaring the inverse transform of $\mathbf{1}_{[-1/2,1/2]}$ and convolving that interval with itself. Since ϕ is not Schwartz, we do not pair it with an arbitrary tempered measure before proving integrability. Instead put

$$\phi_\varepsilon(x) = e^{-\pi\varepsilon x^2}\phi(x), \qquad G_\varepsilon(\xi) = \varepsilon^{-1/2}e^{-\pi\xi^2/\varepsilon}.$$

For $\varepsilon > 0$ these are Schwartz functions and $\widehat{\phi_\varepsilon} = T * G_\varepsilon \geq 0$. Self-duality gives

$$\int \phi_\varepsilon\,d\mu = \int (T * G_\varepsilon)\,d\mu. \tag{3}$$

The right side tends to $\int T\,d\mu = 1$. Here are the needed tail details. A positive tempered measure has polynomial mass growth: testing against a nonnegative Schwartz function that is at least one on $[-1,1]$ and dilating it bounds $\mu([-R,R])$ by a fixed polynomial in $R \geq 1$. On compact sets $0 \leq T * G_\varepsilon \leq 1$ and pointwise convergence to the continuous function T holds, including at ± 1. For $|\xi| \geq 2$ and $0 < \varepsilon \leq 1$, integration over $[-1,1]$ and $|\xi - y| \geq |\xi| - 1$ give

$$(T * G_\varepsilon)(\xi) \leq 2\varepsilon^{-1/2}e^{-\pi(|\xi|-1)^2/\varepsilon} \leq 2e^{-\pi(|\xi|-1)^2}.$$

The last inequality follows by maximizing $v^{1/2}e^{-\pi d^2 v}$ for $v \geq 1, d \geq 1$. This Gaussian envelope is integrable against a measure of polynomial growth. Dominated convergence therefore applies to the right side of (3).

On the left, $\phi_\varepsilon \uparrow \phi$ as $\varepsilon \downarrow 0$. Monotone convergence proves $\int \phi\,d\mu = 1$, now including finiteness. The atom at zero already contributes one. Positivity forces all other mass into the zeros of ϕ, namely the nonzero integers. Thus $\operatorname{supp}\mu \subset \mathbb{Z}$. Because $\widehat{\mu} = \mu$ is a measure supported on $\mathbb{Z}$, $(e^{2\pi i\xi} - 1)\widehat{\mu} = 0$. The Fourier translation rule then implies that μ is invariant under translation by one. All integer atoms therefore have the same mass, equal to one. Rescaling proves the final claim. □

3.1 Decisive mutations and an exact local obstruction

The gap cannot be omitted. The positive measure

$$\mu_* = \frac{2}{3}\delta_{2\mathbb{Z}} + \frac{1}{3}\delta_{\mathbb{Z}/2}$$

has $\mu_*(\{0\}) = 1$ and is self-dual, because $\widehat{\delta_{a\mathbb{Z}}} = a^{-1}\delta_{\mathbb{Z}/a}$. It has an atom of weight $1/3$ at $1/2$, so it is not the integer comb. Its positive-frequency Dirichlet series is

$$Z_*(s) = \frac{2^s + 2^{1-s}}{3}\zeta(s).$$

The multiplier is reflection invariant and equals one at $s = 1$; this control retains residue and completion. It is not a Beurling word source with all primitive values greater than one, since it has frequencies below one. Conversely, $\delta_{\mathbb{Z}} + \delta_{3/2} + \delta_{-3/2}$ retains positivity, normalization and the gap but fails self-duality by the proof above.

Suppose a positive unit-gap measure has atoms of mass at least $w > 0$ at $\pm x$, where $x \geq 1$ is not an integer. Write $d = \operatorname{dist}(x, \mathbb{Z}) \in (0, 1/2]$. Concavity of sine on $[0, \pi/2]$ gives $\sin(\pi d) \geq 2d$, hence

$$\pi^2 \left(\int \phi \, d\mu - \mu(\{0\}) \right) \geq \frac{8wd^2}{x^2} > 0, \tag{4}$$

with an infinite left side allowed. This contradicts self-duality without an infinite numerical sum. The counterfeit atom $x = 9/2, w = 1$ already gives the exact pair contribution $8/81$ to the left side. The rational code certifies this lower bound; it does not certify the Fourier identity by a finite sample.

4 What an infinite-source completion must actually supply

Let B be a locally finite multiset in $[1, \infty)$, with multiplicities $m_b \in \mathbb{N}$ and $m_1 = 1$. A Beurling word multiset is a special case: take primitive real bases greater than one, finitely many at most each bound, and count all their finite commutative words, including the unit. Assume $Z_B(s) = \sum_{b \in B} m_b b^{-s}$ converges absolutely in some right half-plane, where the sum here and below is over distinct values. No Euler product is required for the first conclusion of the next theorem.

Theorem 4 (A sufficient infinite-source rigidity contract)**.** *Define $\Lambda_B(s) = \pi^{-s/2}\Gamma(s/2)Z_B(s)$ in its initial half-plane. Suppose that for some $c > 1$ in that half-plane:*

1. *Λ_B extends meromorphically to $\mathbb{C}$, with exactly two simple poles, at $0, 1$, with residues $-\rho, \rho$, where $\rho > 0$;*
2. *$\Lambda_B(s) = \Lambda_B(1 - s)$;*
3. *uniformly for $1 - c \leq \sigma \leq c$ as $|t| \to \infty$, $|\Lambda_B(\sigma + it)| = O((1 + |t|)^{-2})$.*

Then $\rho = 1$ and B is the positive integer multiset, each integer having multiplicity one. If B is a Beurling word multiset, its primitive multiset is exactly the ordinary primes, with multiplicity one.

Proof. Absolute convergence at c implies $\sum_{b \leq X} m_b \leq X^c Z_B(c)$, so the symmetric source measure below is tempered and all Gaussian sums converge. Put $f(t) = \sum_b m_b e^{-\pi b^2 t}$ for $t > 0$. Mellin inversion of the exponential, followed by absolutely justified summation, gives

$$f(t) = \frac{1}{4\pi i} \int_{\Re s = c} \Lambda_B(s) t^{-s/2} \, ds. \tag{5}$$

The absolute justification on this line follows from $\sum_b m_b b^{-c} < \infty$ and the integrability of $\Gamma((c + iy)/2)$ in y. Shift the contour to $\Re s = 1 - c$. The uniform decay hypothesis makes the horizontal

edges vanish and both vertical integrals absolutely convergent. The residues at 1 and 0 give respectively $\rho t^{-1/2}/2$ and $-\rho/2$. Substituting $s = 1 - w$ in the remaining vertical integral and using reflection yields

$$f(t) = t^{-1/2} f(1/t) + \frac{\rho}{2}(t^{-1/2} - 1).$$

Consequently $\theta(t) = \rho + 2f(t)$ obeys $\theta(t) = t^{-1/2}\theta(1/t)$.

Define the positive even tempered measure

$$\mu = \rho\delta_0 + \sum_b m_b(\delta_b + \delta_{-b}).$$

For $g_t(x) = e^{-\pi t x^2}$, the Gaussian transform formula and theta reciprocity imply $\langle \mu - \widehat{\mu}, g_t \rangle = 0$ for every $t > 0$. The even-distribution separation argument in Lemma 5 below gives $\mu = \widehat{\mu}$. Theorem 3, after rescaling by ρ, gives $\mu = \rho\delta_{\mathbb{Z}}$. The source atom at 1 has mass $m_1 = 1$, so $\rho = 1$. Thus $B = \mathbb{N}_{>0}$ with multiplicity one.

For a Beurling word source, each primitive must now be an integer at least two. Each ordinary prime must occur as a primitive, since a product of integers at least two can equal a prime only when it is a single factor. A repeated primitive or any composite primitive would give at least two different words at the same integer (using ordinary prime factors), contradicting multiplicity one. This identifies the primitive multiset. □

Lemma 5 (Centered Gaussians separate even tempered distributions)**.** *If $T \in \mathcal{S}'(\mathbb{R})$ is even and $\langle T, e^{-\pi t x^2} \rangle = 0$ for every $t > 0$, then $T = 0$.*

Proof. The map $t \mapsto e^{-\pi t x^2}$ is smooth with values in Schwartz space on $t > 0$. Differentiating at $t = 1$ shows that T annihilates $x^{2k} e^{-\pi x^2}$ for every $k \geq 0$. It thus annihilates every even dilated Hermite function with Gaussian factor $e^{-\pi x^2}$, using the oscillator $-d^2/dx^2 + 4\pi^2 x^2$. Its polynomial factor is even. The corresponding dilated Hermite expansions of Schwartz functions converge in the Schwartz topology, so these functions span a dense subspace of the even Schwartz functions. For completeness, the topology assertion follows by applying powers of the harmonic oscillator: the Hermite coefficients of a Schwartz function decay faster than every inverse power of the index, while every Schwartz seminorm of a Hermite function grows at most polynomially, by the raising and lowering identities. Therefore the expansion and its derivatives with polynomial weights converge in every Schwartz seminorm. Continuity of T shows that it annihilates all even Schwartz functions. Its evenness makes it annihilate odd Schwartz functions as well, proving $T = 0$. □

Analytic boundary. The strip decay in Theorem 4 is a strong, sufficient condition chosen to make every contour step explicit. It is not deduced from the displayed functional equation, residue data, finite Euler jets or finite observations. Absolute convergence on a right half-plane suffices for polynomial source growth, so a separate density-one hypothesis is not needed. The theorem is not a classification without growth conditions, nor a claim that every generalized prime zeta has the ordinary completion. It excludes all nonordinary positive Beurling word sources satisfying its complete contract; it does not establish that contract from the repository's observer axioms.

5 Prior art, status and the next falsifier

Hilberdink and Lapidus [1] develop generalized prime zeta functions and relate Mellin functional equations to modular identities under explicit analytic hypotheses (Theorem 3.2). Their paired

completions need not be the same Gaussian completion. This supplies context, not a classification premise for Theorem 4.

Burnol [2] recalls Hamburger's converse for ordinary integer Dirichlet series and gives extensions, including generalized exponents. One cannot apply the ordinary-support theorem to reconstruct integer support by assuming that support at the outset. Our positive-gap proof makes that step explicit under the stronger contour contract stated here.

Baake, Spindeler and Strungaru [3] study Fourier eigenmeasures far beyond a single lattice and record the lattice Poisson formula and Hermite framework. Self-duality alone is therefore not our identifying premise. The added positive unit gap yields an extremal sinc argument of classical Fourier type; priority for this particular formulation is unverified.

All three theorems have complete proofs above and exact adverse controls. Their status is *proved locally, scholarly novelty unverified*. The finite tests exercise arithmetic identities and witnesses, not universal analytic claims. The cheapest falsifier for finite camouflage is (2); for a noninteger positive source atom it is (4). The remaining source lemma is to derive the full normalized Gaussian-completed continuation and strip control from independent source axioms, or to exhibit a nonordinary source that passes those axioms while violating one of these analytic hypotheses. Merely renaming reflection as a source law does not close that dependency.

References

[1] T. Hilberdink and M. L. Lapidus, *Beurling Zeta Functions, Generalised Primes, and Fractal Membranes*, Acta Appl. Math. 94 (2006), 21–48. doi:10.1007/s10440-006-9063-0. Primary full text: `https://arxiv.org/abs/math/0410270`.

[2] J.-F. Burnol, *Deux extensions de Théorèmes de Hamburger*, 2011, version 2. `https://arxiv.org/abs/1106.4749`.

[3] M. Baake, T. Spindeler and N. Strungaru, *On Eigenmeasures Under Fourier Transform*, J. Fourier Anal. Appl. 29 (2023), article 65. doi:10.1007/s00041-023-10045-z.